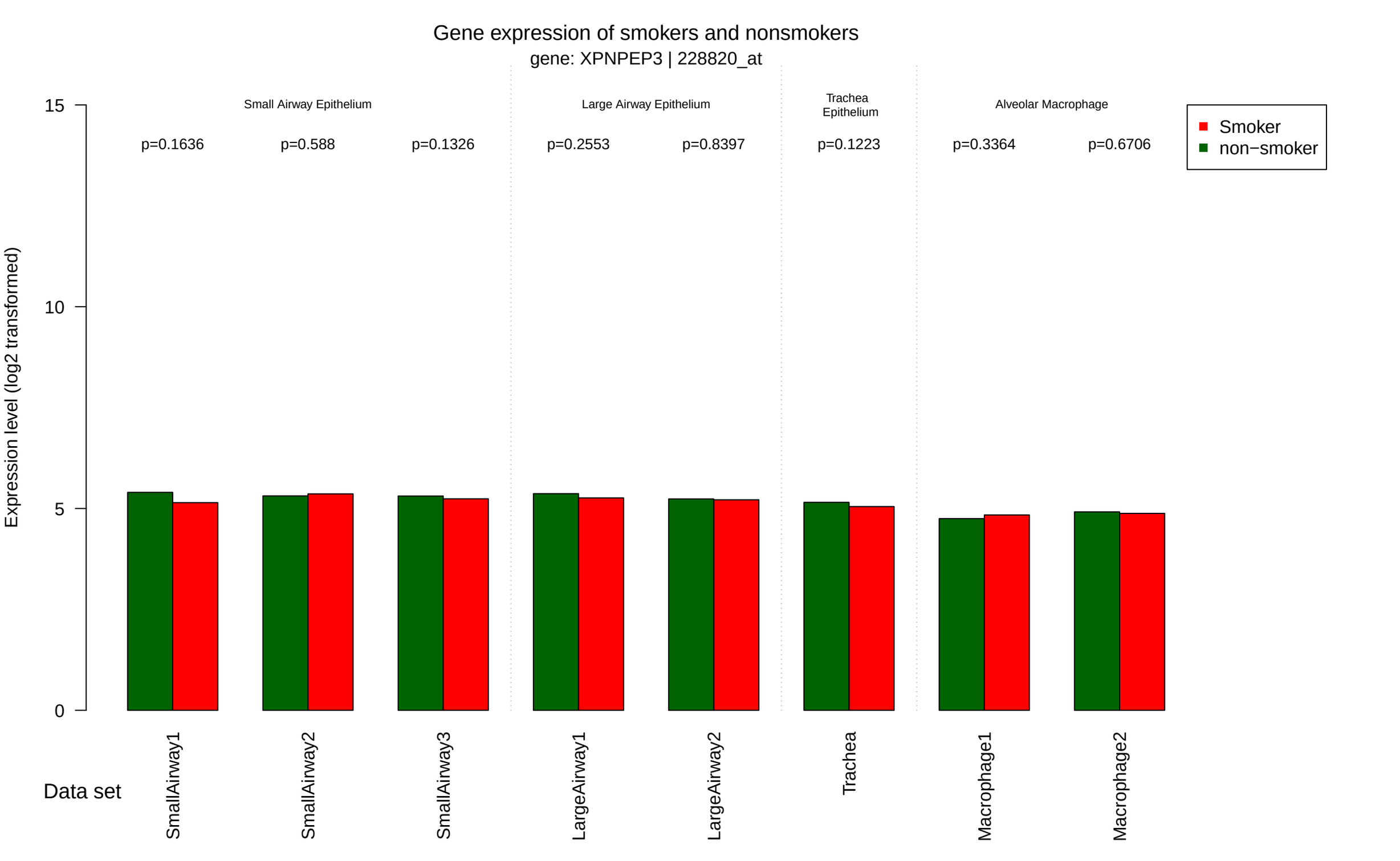
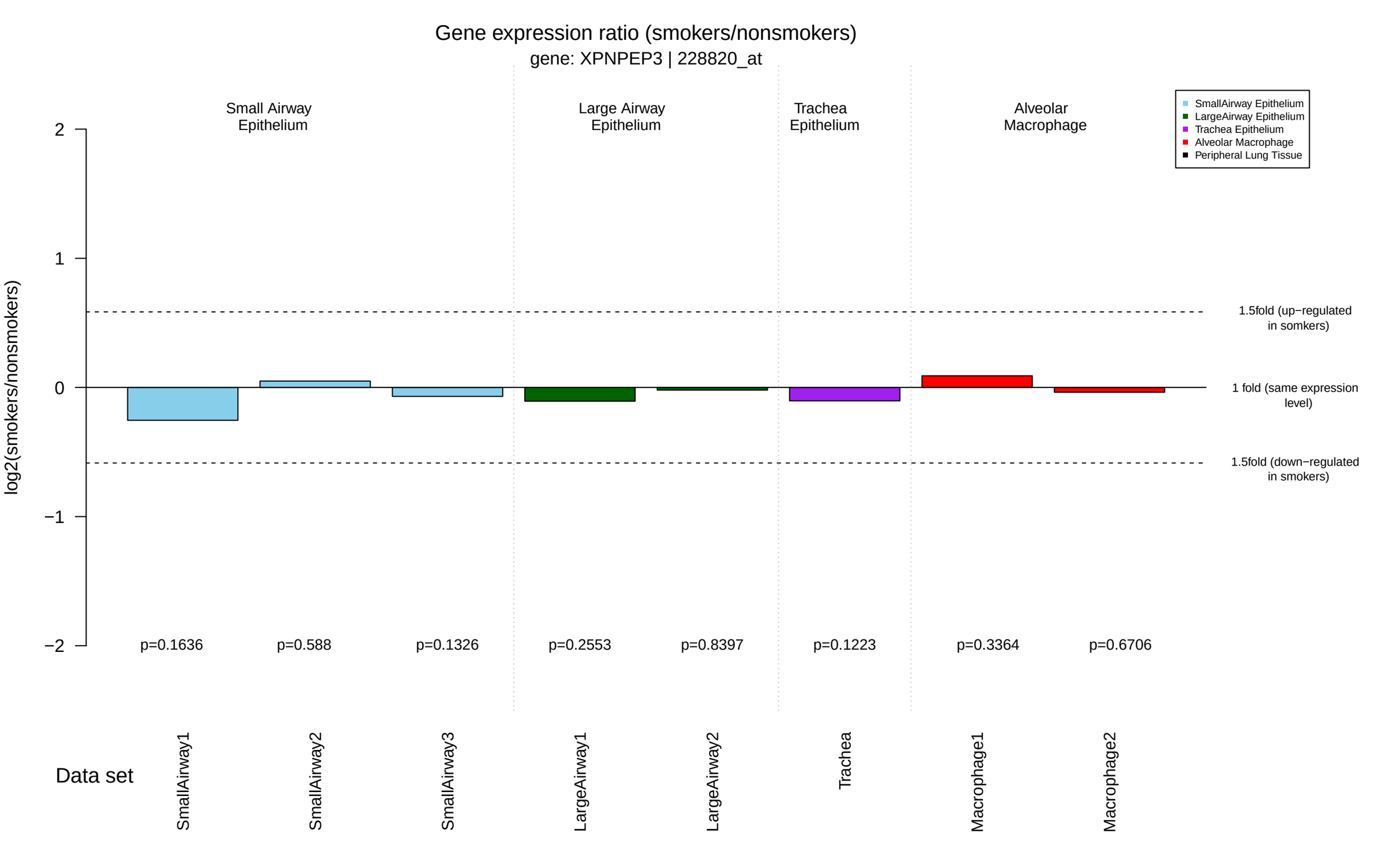
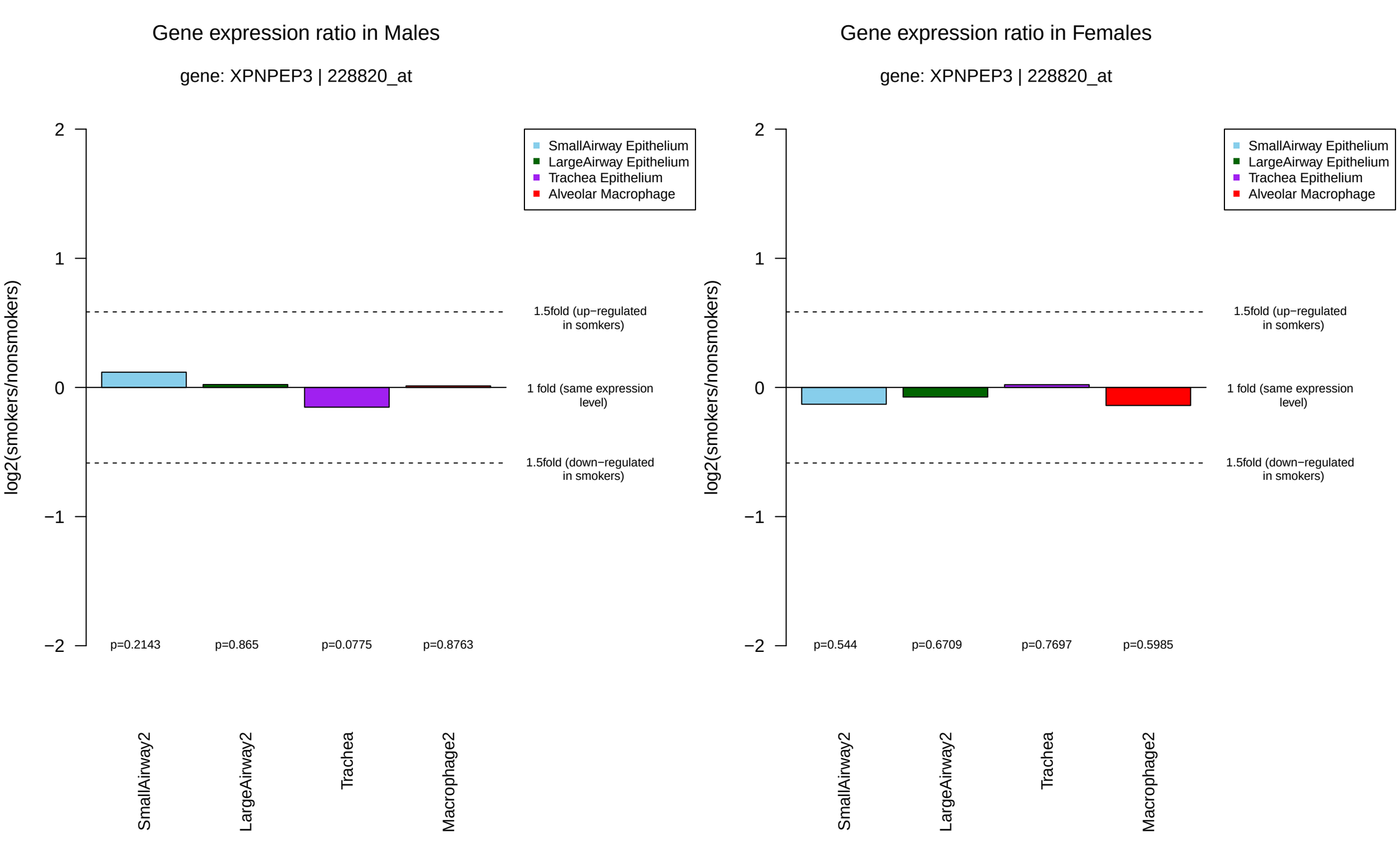
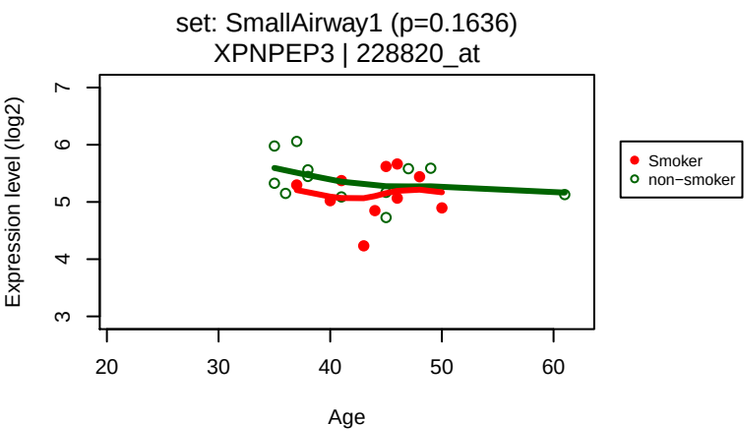
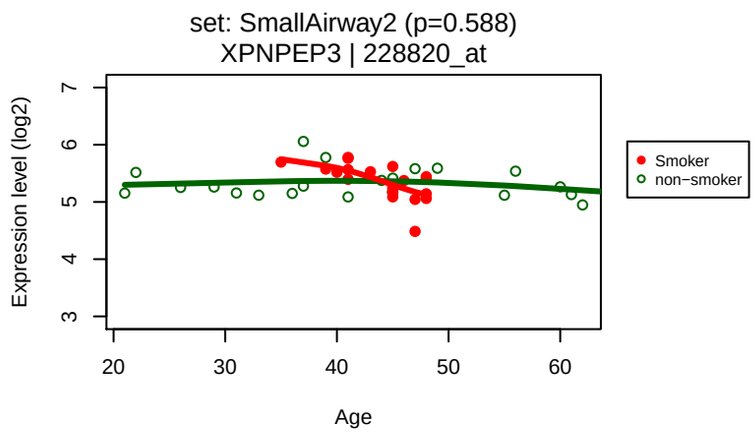
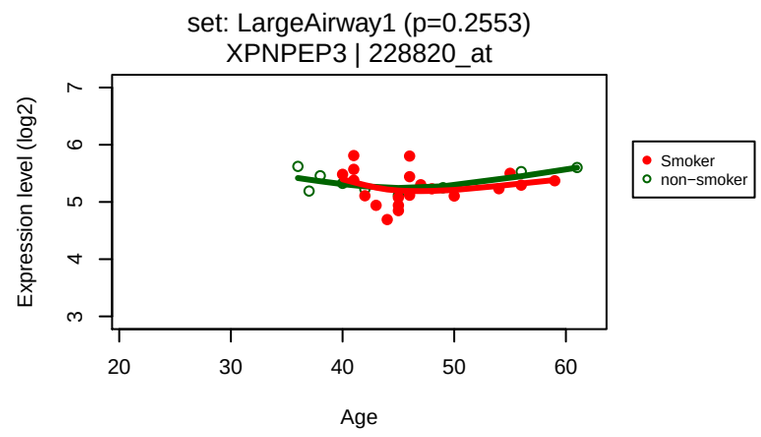

Gene expression of smokers and nonsmokers
gene: XPNPEP3 | 228820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1636
p=0.588
p=0.1326
p=0.2553
p=0.8397
p=0.1223
p=0.3364
p=0.6706
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

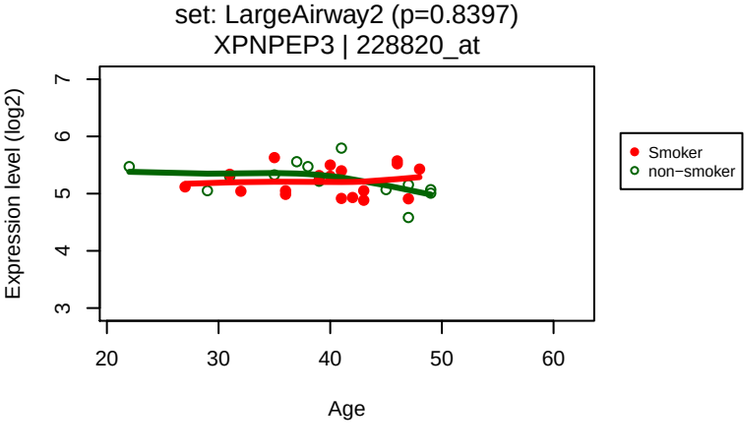

Gene expression ratio (smokers/nonsmokers)
gene: XPNPEP3 | 228820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1636
p=0.588
p=0.1326
p=0.2553
p=0.8397
p=0.1223
p=0.3364
p=0.6706
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

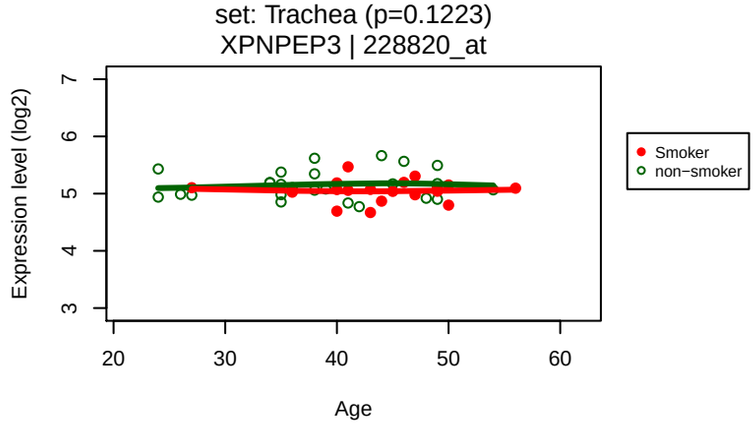

Gene expression ratio in Males
gene: XPNPEP3 | 228820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2143
p=0.865
p=0.0775
p=0.8763
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XPNPEP3 | 228820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.544
p=0.6709
p=0.7697
p=0.5985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

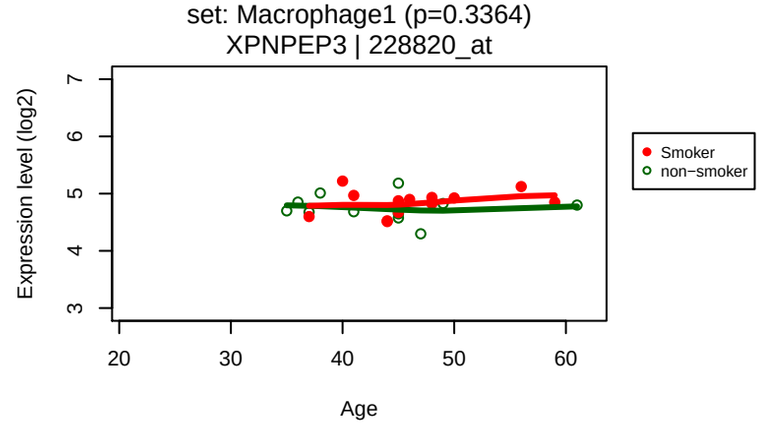

set: SmallAirway1 (p=0.1636)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker

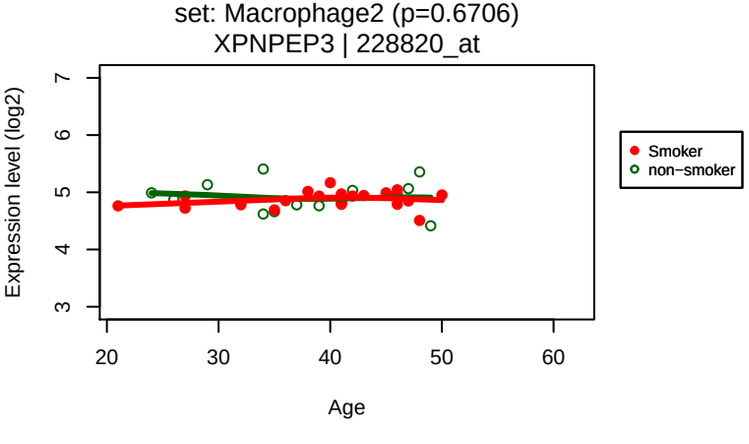

set: SmallAirway2 (p=0.588)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker

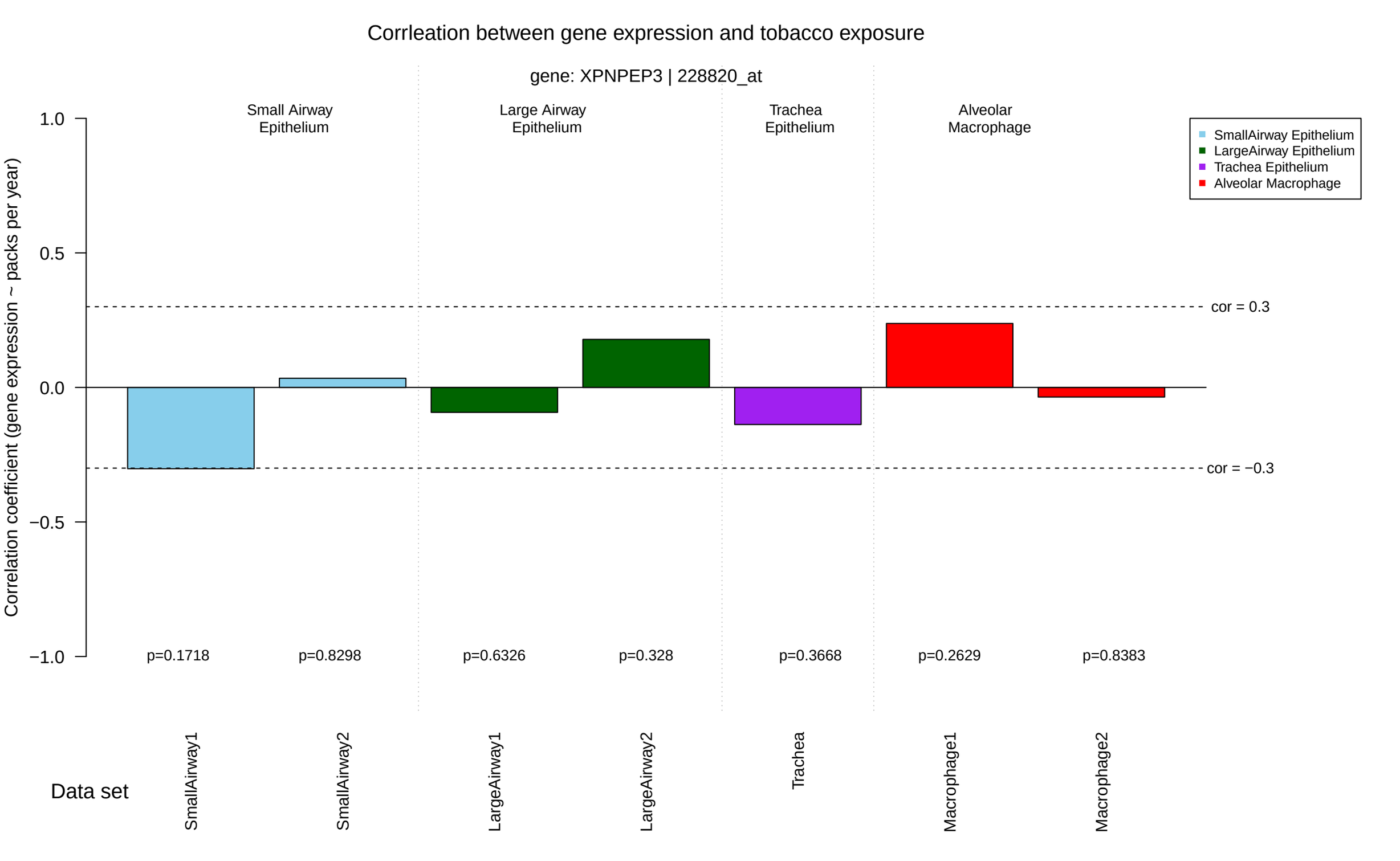

set: LargeAirway1 (p=0.2553)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8397)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1223)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3364)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6706)
XPNPEP3 | 228820_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XPNPEP3 | 228820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1718
p=0.8298
p=0.6326
p=0.328
p=0.3668
p=0.2629
p=0.8383
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2